

KELT-23Ab

R-2162

Benchmark run · workflow W8-benchmark-deep · record deep-bench_KELT-23A_250324 · created 2026-08-12T23:51:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460758.77489 +/- 0.00095 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.129 +/- 0.014
Transit depth (Rp/Rs)^2	1.66 +/- 0.35 [%]
Orbital Inclination (inc)	85.83 +/- 0.51
Airmass coefficient 1 (a1)	1.2273 +/- 0.0057
Airmass coefficient 2 (a2)	-0.154 +/- 0.017
Scatter in the residuals of the lightcurve fit is	0.46 %
Best Comparison Star	None
Optimal Aperture	4.02
Optimal Annulus	9.55
Transit Duration (day)	0.0605 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.129 $\pm$ 0.014 vs 0.132 (deviation 0.21 σ)
Duration fit vs published	0.0605 d vs 0.0954 d (deviation 16.63 σ)
Mid-time O-C	-1.5 min
Depth SNR	9.2
Residual scatter	0.46 %

Light curve

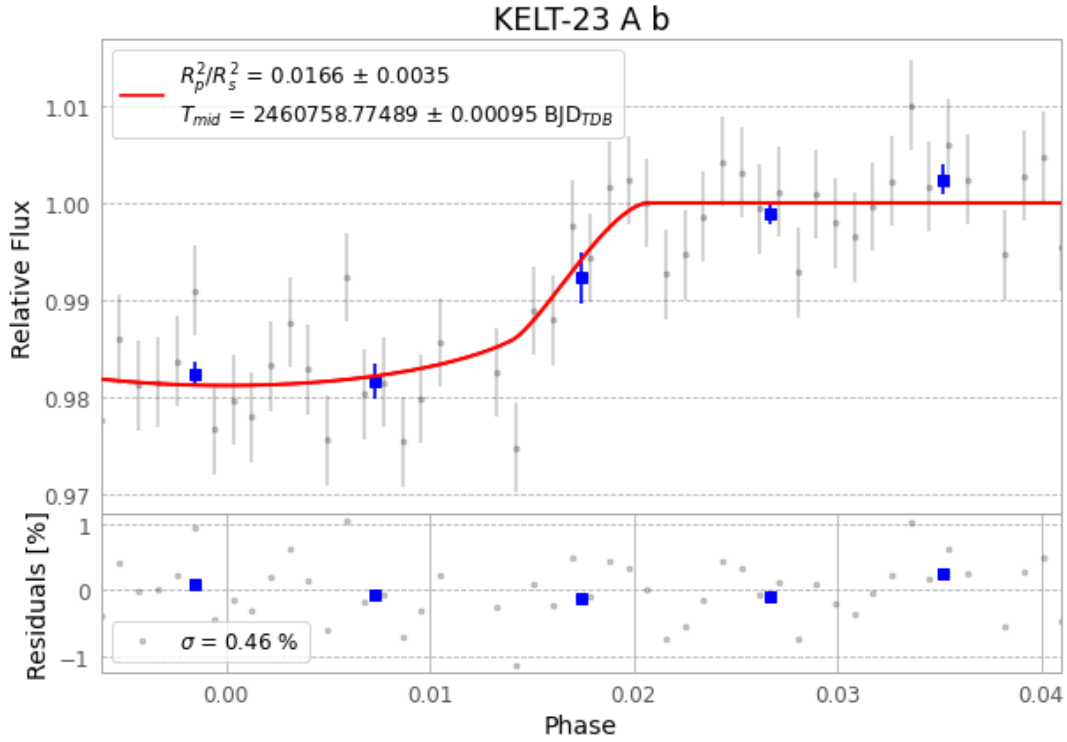


Figure 1 — FinalLightCurve_KELT-23 A b_2025-03-23.png (SHA-256 68bc6d0c37d4dc4f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 198], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0659e1c3a136a112...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

52 FITS frames (34 MB), KELT-23A250324061216.FITS → KELT-23A250324084515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-KELT-23A-250324.log (80edeac76607...), FinalLightCurve_KELT-23 A b_2025-03-23.png (68bc6d0c37d4...), FinalLightCurve_KELT-23 A b_2025-03-23.csv (c3f14ba8535e...)

Compute resources

Wall time 96.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 23:51Z.